Number 10 — Rational and irrational numbers · Surds

No calculator. Marks in brackets. ★ = harder question. Leave answers as exact surds in their simplest form.

1 Simplify [4]

a $\sqrt{80} - \sqrt{20}$ **b** $\sqrt{\dfrac{81}{49}}$

a $\sqrt{80} = \sqrt{16 \times 5} = 4\sqrt{5}$ and $\sqrt{20} = 2\sqrt{5}$, so $4\sqrt{5} - 2\sqrt{5} = 2\sqrt{5}$. **b** $\dfrac{\sqrt{81}}{\sqrt{49}} = \dfrac{9}{7}$.

2 Simplify [4]

a $\sqrt{18} + 3\sqrt{8}$ **b** $2\sqrt{75} - \sqrt{27}$

a $3\sqrt{2} + 3 \times 2\sqrt{2} = 3\sqrt{2} + 6\sqrt{2} = 9\sqrt{2}$. **b** $2 \times 5\sqrt{3} - 3\sqrt{3} = 10\sqrt{3} - 3\sqrt{3} = 7\sqrt{3}$.

3 Expand and simplify [4]

a $(3 - \sqrt{2})^2$ **b** $(2 + \sqrt{5})(2 - \sqrt{5})$

a $9 - 6\sqrt{2} + 2 = 11 - 6\sqrt{2}$. **b** Difference of two squares: $4 - 5 = -1$ (the surd terms cancel).

4 Rationalise the denominator and simplify [4]

a $\dfrac{2 + \sqrt{3}}{\sqrt{3}}$ **b** $\dfrac{5}{3 - \sqrt{7}}$

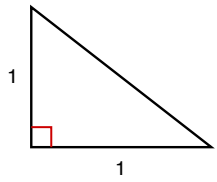
a Multiply top and bottom by $\sqrt{3}$: $\dfrac{2\sqrt{3} + 3}{3}$. **b** Multiply top and bottom by $3 + \sqrt{7}$: $\dfrac{5(3 + \sqrt{7})}{9 - 7} = \dfrac{5(3 + \sqrt{7})}{2} = \dfrac{15 + 5\sqrt{7}}{2}$.

5 Work out, giving each answer in the form $\dfrac{a\sqrt{2}}{b}$: [4]

a $\dfrac{5}{\sqrt{2}} + \dfrac{3}{\sqrt{8}}$ **b** $\dfrac{10}{\sqrt{50}} + \dfrac{3}{\sqrt{2}}$

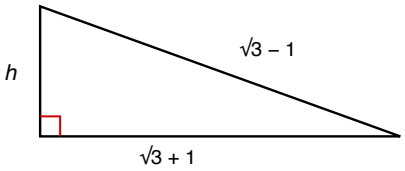
a $\sqrt{8} = 2\sqrt{2}$, so $\dfrac{5}{\sqrt{2}} + \dfrac{3}{2\sqrt{2}} = \dfrac{10 + 3}{2\sqrt{2}} = \dfrac{13}{2\sqrt{2}} = \dfrac{13\sqrt{2}}{4}$. **b** $\sqrt{50} = 5\sqrt{2}$, so $\dfrac{10}{5\sqrt{2}} + \dfrac{3}{\sqrt{2}} = \dfrac{2}{\sqrt{2}} + \dfrac{3}{\sqrt{2}} = \dfrac{5}{\sqrt{2}} = \dfrac{5\sqrt{2}}{2}$.

6 The diagram shows an isosceles right-angled triangle with two sides of length 1. **a** Use it to write $\sin 45°$ as a surd in its simplest form. **b** A boat sails 24 km on a bearing of 045°. How far east has it travelled? Give an exact answer. [2]



> **a** Hypotenuse $= \sqrt{1+1} = \sqrt{2}$, so $\sin 45° = \dfrac{1}{\sqrt{2}} = \dfrac{\sqrt{2}}{2}$. **b** East $= 24\sin 45° = 24 \times \dfrac{\sqrt{2}}{2} = 12\sqrt{2}$ km.

7 A right-angled triangle has shorter sides $\sqrt{3}+1$ and $\sqrt{3}-1$. Find the exact length h of the hypotenuse. [3]



> Pythagoras: $h^2 = (\sqrt{3}+1)^2 + (\sqrt{3}-1)^2 = (3+2\sqrt{3}+1) + (3-2\sqrt{3}+1) = 8$, so $h = \sqrt{8} = 2\sqrt{2}$.

Total: 25 marks

Check: $\sqrt{ab} = \sqrt{a}\sqrt{b}$ — pull out the largest square factor · add/subtract only like surds · $(a+\sqrt{b})(a-\sqrt{b}) = a^2 - b$ · rationalise $\frac{1}{\sqrt{b}}$ with $\sqrt{b}$, and $\frac{1}{a\pm\sqrt{b}}$ with $a \mp \sqrt{b}$ · a surd answer is exact; a decimal is not.